

# Investigating Self-Supervised Autoencoder Learning for VideoQA

Phil Gailinas

ECE-551 – Problems in Machine Learning

Instructor: Dr. Marios Pattichis

University of New Mexico

pgailinas@unm.edu

**Abstract**—Video Question Answering (VideoQA) requires joint reasoning over visual content and natural language. This paper compares self-supervised autoencoder, pretrained CLIP, and hybrid CLIP–autoencoder video representations within a common representation-based VideoQA framework, together with direct multimodal inference using Qwen2-VL. All representation-based approaches share identical text representations, prediction models, and evaluation procedures, isolating the influence of video representation quality. Experiments on the NExT-QA benchmark show that Qwen2-VL achieved the highest accuracy (79.84%), while pretrained CLIP provided the strongest representation-based performance (46.42%). Although the hybrid representation improved upon the autoencoder alone (31.29% versus 21.78%), it did not surpass pretrained CLIP. The results indicate that semantic representation quality, rather than reconstruction quality alone, is the primary factor governing VideoQA performance.

**Index Terms**—Video Question Answering, Representation Learning, Self-Supervised Learning, Autoencoder, CLIP, Vision-Language Models, Multimodal Learning, Semantic Alignment

## I. INTRODUCTION

Video Question Answering (VideoQA) is a multimodal learning task requiring joint reasoning over visual content and natural language, including spatial, temporal, and causal relationships within video sequences [1].

Recent vision-language foundation models, such as Qwen2-VL, have achieved strong VideoQA performance through end-to-end multimodal reasoning [2]. In contrast, representation-based approaches generate reusable feature embeddings. CLIP learns semantically aligned vision-language representations [3], whereas reconstruction-based autoencoders learn latent representations by optimizing visual reconstruction [4], [5]. Whether reconstruction-based representations alone preserve the semantic information required for VideoQA remains an open question [7].

This work investigates that question through a controlled comparison of three VideoQA pipelines using the NExT-QA benchmark. A common representation-based framework evaluates pretrained CLIP, self-supervised autoencoder, and hybrid CLIP–autoencoder video representations while holding text representations, downstream prediction models, and evaluation procedures constant.

The primary contributions of this work are as follows:

- A reproducible framework for comparing foundation-model inference with representation-based VideoQA approaches.
- A controlled comparison of Qwen2-VL, pretrained CLIP, hybrid CLIP–autoencoder, and self-supervised autoencoder representations using NExT-QA.
- Experimental evidence that semantic representation quality has a greater influence on VideoQA performance than classifier complexity.
- Identification of semantic alignment as a promising direction for improving reconstruction-based video representations.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the NExT-QA benchmark dataset. Section IV presents the methodology. Section V describes the experimental setup. Section VI presents the results. Section VII discusses the findings. Section VIII outlines future work, and Section IX concludes the paper.

## II. RELATED WORK

VideoQA requires joint understanding of visual content, language, and temporal relationships. The NExT-QA benchmark emphasizes temporal and causal reasoning through multiple-choice questions associated with real-world videos [1].

Large vision-language models have substantially advanced multimodal reasoning. Qwen2-VL performs end-to-end multimodal reasoning through a unified transformer architecture [2]. In contrast, representation-based approaches rely on reusable feature embeddings. CLIP learns a shared semantic embedding space through large-scale vision-language contrastive learning [3].

Reconstruction-based self-supervised learning provides an alternative approach to representation learning. Masked Autoencoders demonstrated the effectiveness of reconstruction objectives [4], and similar methods have been extended to video representation learning [5]. ReasVQA further highlights the importance of semantically meaningful latent representations for higher-level VideoQA reasoning [6].

Although prior work has explored foundation models, semantic representation learning, and reconstruction-based learning, few studies have directly compared pretrained,

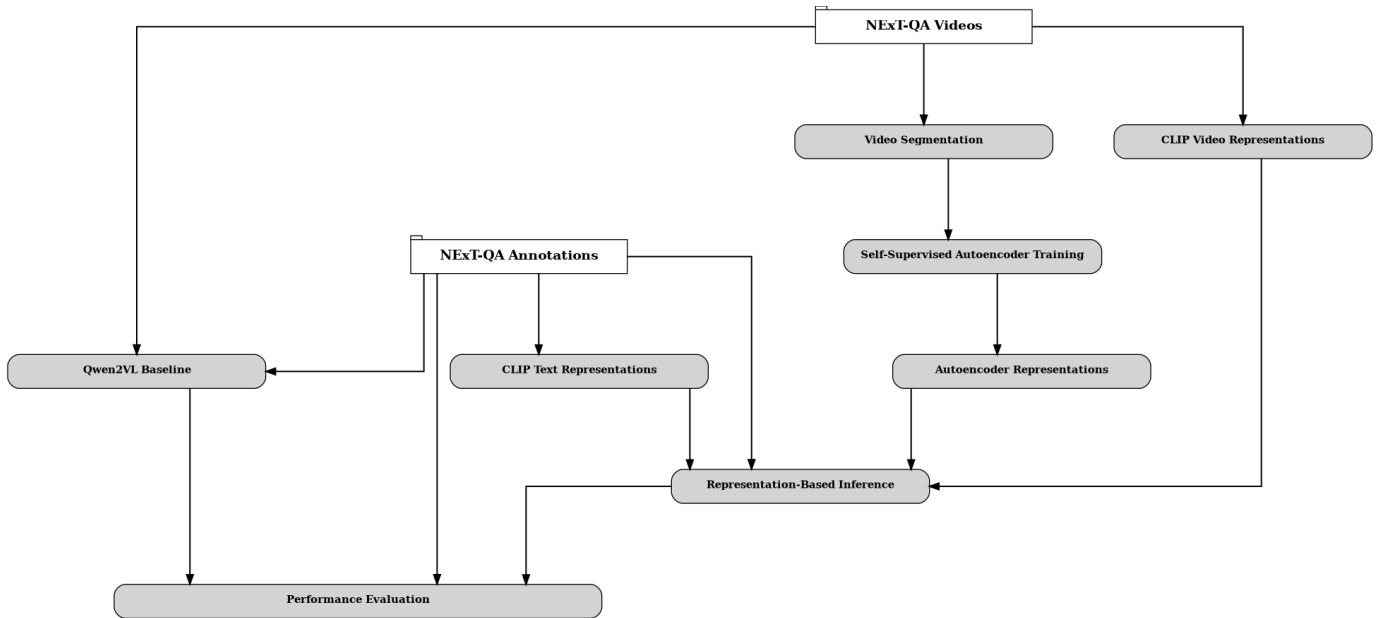


Fig. 1. Overall methodology comparing the Qwen2-VL baseline with the representation-based VideoQA framework using pretrained CLIP, self-supervised autoencoder, and hybrid video representations.

reconstruction-based, and hybrid video representations within a common VideoQA framework. This study addresses that gap through a controlled comparison that isolates the influence of video representation quality on downstream VideoQA performance [7].

### III. DATASET

All experiments utilize the NExT-QA benchmark dataset, a large-scale VideoQA benchmark designed to evaluate temporal, causal, and descriptive reasoning over realistic video content [1]. Unlike earlier VideoQA datasets that emphasize object recognition or simple event descriptions, NExT-QA requires multiple-choice reasoning about actions, temporal relationships, and cause-and-effect interactions within a video.

Table I summarizes the official NExT-QA dataset partitions used throughout this study.

TABLE I  
SUMMARY OF NExT-QA DATASET

| Split      | Videos | QA Records |
|------------|--------|------------|
| Training   | 3,870  | 34,132     |
| Validation | 570    | 4,996      |
| Test       | 1,000  | 8,564      |
| Total      | 5,440  | 47,692     |

The complete NExT-QA dataset contains 5,440 videos and 47,692 multiple-choice question-answer records. Each question includes five candidate answers with exactly one correct response. The official training, validation, and testing splits support reproducible evaluation across VideoQA methods.

The official NExT-QA training, validation, and testing splits were preserved throughout this study. Development experi-

ments used deterministic subsets of the training and validation splits to reduce computational cost while maintaining reproducibility. Final experiments used the complete official training and validation splits while preserving the benchmark evaluation protocol.

The three VideoQA pipelines used the NExT-QA videos differently. Qwen2-VL operated directly on the original videos, the CLIP pipeline generated pretrained video representations, and the autoencoder pipeline learned video representations from segmented training videos. Regardless of the representation source, all experiments evaluated the same NExT-QA multiple-choice question-answer pairs.

### IV. METHODOLOGY

This study compares three VideoQA pipelines within a common evaluation framework. Figure 1 summarizes the overall methodology, including the Qwen2-VL baseline and the representation-based framework evaluating pretrained CLIP, self-supervised autoencoder, and hybrid video representations.

#### A. Qwen2-VL Baseline

The first pipeline establishes a reference baseline using the Qwen2-VL-7B vision-language model. The model processes each original video together with the associated question and candidate answers to directly predict the correct response, providing an end-to-end baseline for comparison with the representation-based pipelines.

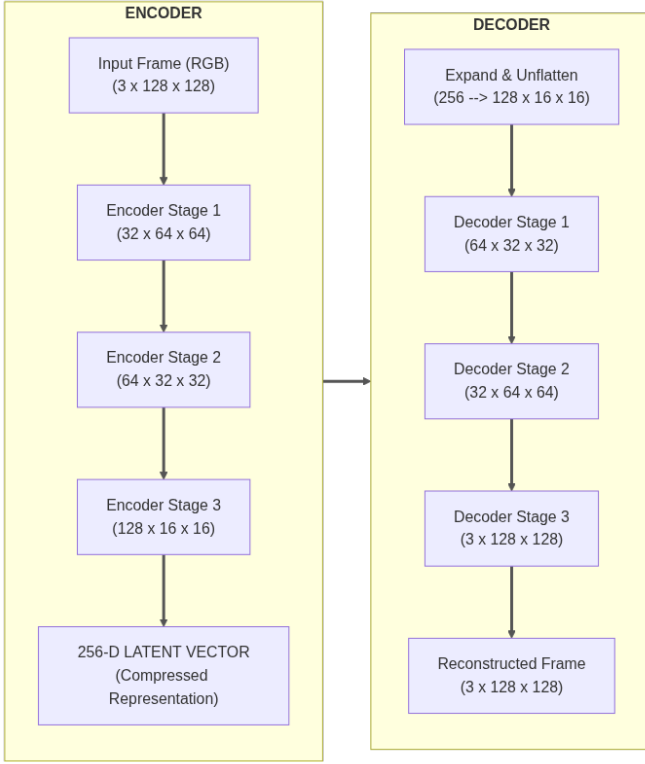


Fig. 2. Convolutional autoencoder used for self-supervised video representation learning.

## B. CLIP Representation Pipeline

The second pipeline adopts a representation-based approach using pretrained CLIP models. Sampled video frames are independently encoded by the pretrained CLIP vision encoder and aggregated into a single fixed-length video representation for each video. Each question-answer candidate pair is encoded using the CLIP text encoder to produce shared text representations.

During VideoQA inference, the corresponding video and question-answer representations are combined using one of five prediction models: Cosine Similarity, Fusion MLP, Interaction Fusion, Gated Fusion, or Bilinear Fusion.

## C. Self-Supervised Autoencoder Pipeline

The third pipeline replaces the pretrained CLIP video representations with representations learned through self-supervised autoencoder training. Videos are first divided into fixed-duration temporal segments. The convolutional autoencoder is trained to reconstruct sampled video frames from compact latent representations without requiring manual annotation or question-answer supervision.

Figure 2 illustrates the convolutional autoencoder used to compress each RGB frame into a 256-dimensional latent representation and reconstruct the original image through a self-supervised reconstruction objective.

Frame reconstruction provides the self-supervised learning objective. After training, the decoder is discarded and the frozen encoder generates latent representations that are aggregated into a single video representation for downstream VideoQA.

Unlike CLIP, the autoencoder is optimized solely through reconstruction loss and receives no explicit semantic supervision. Consequently, successful image reconstruction does not necessarily imply a semantically organized latent space. This study therefore investigates whether reconstruction-based latent representations preserve sufficient semantic information to support downstream VideoQA as effectively as pretrained semantic representations.

## D. Representation-Based VideoQA

All representation sources share a common downstream VideoQA framework in which shared CLIP text representations are combined with pretrained CLIP, autoencoder, or hybrid video representations and evaluated using identical prediction models and evaluation metrics.

For each question, the video representation  $\mathbf{v}$  is compared with the five candidate text representations  $\mathbf{t}_i$ , producing the score vector

$$\mathbf{s} = [s_1, s_2, s_3, s_4, s_5],$$

from which the highest-scoring candidate is selected.

## E. Prediction Models

Five prediction models were evaluated within the common representation-based VideoQA framework. Let  $\mathbf{v}$  denote the video representation,  $\mathbf{t}_i$  the representation of the  $i$ th candidate answer, and  $s_i$  the corresponding prediction score. For each question, the candidate with the highest score is selected as the predicted answer.

**Cosine Similarity** directly compares the video and text representations by measuring the angle between the two vectors,

$$s_i = \text{cosine}(\mathbf{v}, \mathbf{t}_i).$$

Because no trainable parameters are used, Cosine Similarity provides a simple baseline that evaluates the semantic alignment of the learned representations.

**Fusion MLP** first concatenates the video and text representations into a single feature vector,

$$\mathbf{x}_i = [\mathbf{v}; \mathbf{t}_i],$$

which is then processed by a multilayer perceptron (MLP) to predict the candidate score. The network learns nonlinear relationships between the video and text features during supervised training.

**Interaction Fusion** extends the Fusion MLP by including additional features that explicitly capture differences and similarities between the two representations,

$$\mathbf{x}_i = [\mathbf{v}; \mathbf{t}_i; |\mathbf{v} - \mathbf{t}_i|; \mathbf{v} \odot \mathbf{t}_i].$$

The absolute difference highlights feature disagreement, while the element-wise product emphasizes feature agreement.

**Gated Fusion** learns how much information to retain from each modality by computing a feature-wise gate,

$$\mathbf{x}_i = \mathbf{g}_i \odot \mathbf{v} + (1 - \mathbf{g}_i) \odot \mathbf{t}_i,$$

where  $\mathbf{g}_i$  contains learned values between zero and one. Larger values emphasize video features, while smaller values emphasize text features.

**Bilinear Fusion** models multiplicative interactions between the video and text representations,

$$s_i = \mathbf{v}^T W \mathbf{t}_i,$$

where  $W$  is a learned weight matrix. Unlike simple concatenation, Bilinear Fusion directly learns interactions between individual video and text features, allowing richer multimodal relationships to be modeled.

## V. EXPERIMENTAL SETUP

All experiments used the official NExT-QA benchmark, identical multiple-choice question-answer pairs, and a common evaluation protocol, allowing performance differences to be attributed primarily to the video representations.

The representation-based experiments evaluated three video representation sources: pretrained CLIP embeddings, self-supervised autoencoder embeddings learned from the NExT-QA training split, and a hybrid representation formed by concatenating normalized 512-dimensional CLIP and 256-dimensional autoencoder embeddings. Five prediction models were evaluated using a common framework: Cosine Similarity as a parameter-free baseline and four learned multimodal fusion models trained with grouped five-choice cross-entropy classification. Table II summarizes the representation learning configuration.

TABLE II  
REPRESENTATION CONFIGURATION

| Parameter                      | Value     |
|--------------------------------|-----------|
| CLIP Model                     | ViT-B/32  |
| Text Embedding Dimension       | 512       |
| CLIP Video Embedding Dimension | 512       |
| Autoencoder Latent Dimension   | 256       |
| Frame Resolution               | 128 × 128 |
| Frames per Segment             | 8         |
| Segment Duration               | 6 seconds |
| Autoencoder Training Epochs    | 3         |
| Batch Size                     | 8         |
| Learning Rate                  | 0.001     |
| Optimizer                      | Adam      |

Performance was evaluated using multiple-choice classification accuracy on the official NExT-QA validation protocol.

TABLE III  
VIDEOQA PERFORMANCE BY PIPELINE AND PREDICTION MODEL

| Pipeline    | Representation   | Prediction          | Acc. (%)     |
|-------------|------------------|---------------------|--------------|
| Qwen2-VL    | Original         | Foundation          | <b>79.84</b> |
| CLIP        | Pre-Trained      | Bilinear            | <b>46.42</b> |
|             |                  | Interaction         | 45.30        |
|             |                  | Cosine              | 44.18        |
|             |                  | Gated               | 41.61        |
|             |                  | Fusion MLP          | 34.21        |
| Hybrid      | CLIP+Autoencoder | Bilinear            | <b>31.29</b> |
| Autoencoder | Self-Supervised  | Fusion MLP (Dev100) | <b>23.46</b> |
|             |                  | Fusion MLP (Dev500) | 21.78        |

Development used deterministic subsets of the training and validation splits for rapid experimentation, while final results used the complete validation split. All experiments were implemented in Python with PyTorch and executed in Google Colaboratory, using NVIDIA L4 GPUs for representation generation and learned fusion models.

## VI. RESULTS

The evaluated VideoQA pipelines were compared using a common experimental framework. Table III summarizes the resulting classification accuracies.

Four trends emerged from the experimental results. Direct multimodal inference using Qwen2-VL achieved the highest accuracy (79.84%), outperforming all representation-based approaches. Among the representation-based methods, pretrained CLIP video representations consistently exceeded both hybrid CLIP–autoencoder and self-supervised autoencoder representations. Bilinear Fusion produced the strongest representation-based performance (46.42%), while the hybrid representation (31.29%) improved upon the autoencoder-only configurations but remained well below pretrained CLIP.

Across the CLIP-based prediction models, Bilinear Fusion achieved the highest accuracy, followed closely by Interaction Fusion (45.30%) and Cosine Similarity (44.18%). Gated Fusion (41.61%) and Fusion MLP (34.21%) produced lower accuracies, indicating that the choice of prediction model influenced performance, although all CLIP-based methods remained substantially below direct multimodal inference.

Increasing the autoencoder development training subset from 100 to 500 videos did not improve accuracy (23.46% versus 21.78%), indicating that additional development training data alone did not improve performance.

These results indicate that reconstruction-based representations alone do not preserve the semantic information required for competitive VideoQA performance, highlighting the importance of semantically aligned video representations.

## VII. DISCUSSION

A particularly interesting finding of this study was the hybrid representation experiment. Although concatenating the 256-dimensional autoencoder representation with the 512-dimensional pretrained CLIP representation increased the

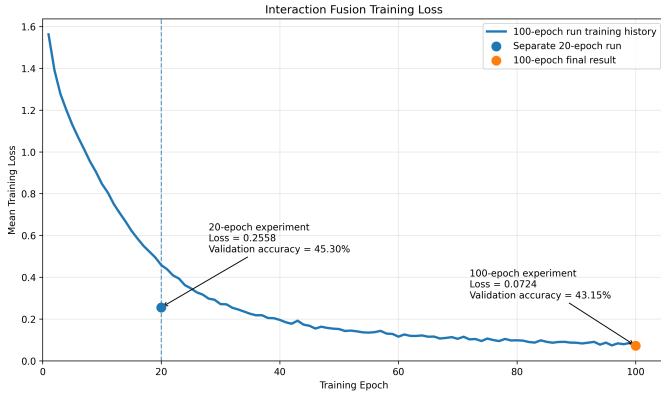


Fig. 3. Interaction Fusion training comparison. Extending training from 20 to 100 epochs reduced training loss but also decreased validation accuracy, indicating overfitting.

amount of representation information available to the prediction model, classification accuracy decreased from 46.42% to 31.29%. This counterintuitive result suggests that representation quality and semantic alignment are more important than representation dimensionality alone and motivates the broader observation that the semantic quality of the learned video representations strongly influences downstream VideoQA performance.

The lower performance of the self-supervised autoencoder reflects the difference between optimizing for visual reconstruction and semantic understanding. While the autoencoder learned compact latent representations sufficient for frame reconstruction, it was not explicitly trained to preserve the semantic information required for VideoQA. In contrast, CLIP representations are optimized through large-scale vision-language training to align visual and textual concepts within a shared embedding space, enabling more effective semantic reasoning.

To determine whether classifier optimization limited performance, the Interaction Fusion classifier was trained for 100 epochs and compared with the standard 20-epoch configuration. Although extended training reduced the final training loss from 0.2558 to 0.0724, validation accuracy declined from 45.30% to 43.15%, indicating overfitting rather than undertraining (Figure 3). These results suggest that the primary limitation of the representation-based framework lies in the semantic quality of the learned video representations rather than the capacity of the downstream prediction model.

## VIII. FUTURE WORK

The results suggest that future research should focus on improving the semantic organization of self-supervised video representations rather than reconstruction quality alone.

A promising direction is to align the autoencoder latent space with pretrained vision-language embedding spaces. Rather than optimizing reconstruction loss alone, future autoencoders could jointly optimize reconstruction fidelity and

semantic consistency by aligning latent representations with pretrained CLIP embeddings. One approach is a teacher–student framework [8], in which pretrained CLIP representations provide semantic targets during training.

The representation-based evaluation framework developed in this study provides a reproducible platform for evaluating semantic-alignment strategies while maintaining a common dataset, prediction models, and evaluation protocol. Additional research may investigate contrastive objectives, latent-space regularization, transformer-based video encoders, larger latent dimensions, and more sophisticated hybrid fusion methods.

Ultimately, the primary challenge is not learning compact visual representations, but learning compact representations that preserve semantic structure.

## IX. CONCLUSION

This paper investigated self-supervised representation learning for VideoQA through a controlled comparison of pretrained CLIP, self-supervised autoencoder, and hybrid CLIP–autoencoder representations within a common evaluation framework. The results demonstrate that semantically aligned video representations substantially outperform reconstruction-based representations and that classifier optimization cannot compensate for limitations in representation quality.

The reproducible evaluation framework developed in this work provides a foundation for future research on representation learning, while the observed performance gap identifies semantic alignment as a promising direction for improving reconstruction-based video representations.

## References

- [1] J. Xiao, X. Shang, A. Yao, and T.-S. Chua, “NExT-QA: Next Phase of Question-Answering to Explaining Temporal Actions,” in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9777–9786, 2021. doi:10.1109/CVPR46437.2021.00965.
- [2] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin, “Qwen2-VL: Enhancing Vision-Language Model’s Perception of the World at Any Resolution,” *arXiv:2409.12191*, 2024. doi:10.48550/arXiv.2409.12191.
- [3] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning Transferable Visual Models From Natural Language Supervision,” in *Proc. 38th International Conference on Machine Learning (ICML)*, vol. 139, pp. 8748–8763, 2021. doi:10.48550/arXiv.2103.00020.
- [4] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked Autoencoders Are Scalable Vision Learners,” in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009, 2022. doi:10.1109/CVPR52688.2022.01553.
- [5] Z. Lai, S. Liu, A. A. Efros, and X. Wang, “Video Autoencoder: Self-Supervised Disentanglement of Static 3D Structure and Motion,” in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9710–9720, 2021. doi:10.1109/ICCV48922.2021.00959.
- [6] J. Liang, X. Meng, H. Zhang, Y. Wang, J. Wei, and D. Zhao, “ReasVQA: Advancing VideoQA with Imperfect Reasoning Process,” in *Proc. 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Volume 1: Long Papers*, pp. 1696–1709, 2025. doi:10.18653/v1/2025.naacl-long.82.

- [7] J. Li, Y. Kementchedjhieva, C. Fierro, and A. Søgaard, “Do Vision and Language Models Share Concepts? A Vector Space Alignment Study,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1232–1249, 2024. doi:10.1162/tacL\_a\_00698.
- [8] G. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” *arXiv:1503.02531*, 2015. doi:10.48550/arXiv.1503.02531.